ME 224 - Fundamentals of Microscale Flow

Numerical Investigation of Split and Recombination (SAR) Microchannel Mixers.

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Introduction

Microfluidic systems operate in the laminar regime ($Re < 100$), where mixing is dominated by molecular diffusion.

- Parallel streamlines prevent turbulence-driven mixing
- Diffusion is slow → requires long channel lengths
- Mixing efficiency decreases as flow rate increases

Passive Solution: Split and Recombine (SAR) Micromixers

- Fluid streams are repeatedly split and recombined
- Increases interfacial area
- Induces transverse flow & chaotic advection
- Enhances mixing without external energy input

Key Challenge

Mixing performance strongly depends on geometry, which affects:

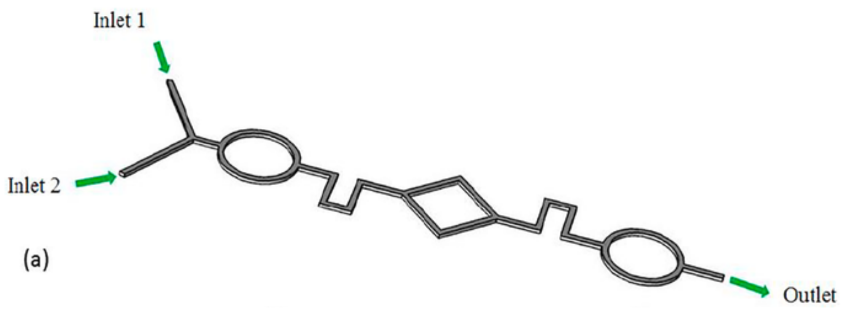
- Secondary flow strength
- Residence time
- Pressure drop penalty

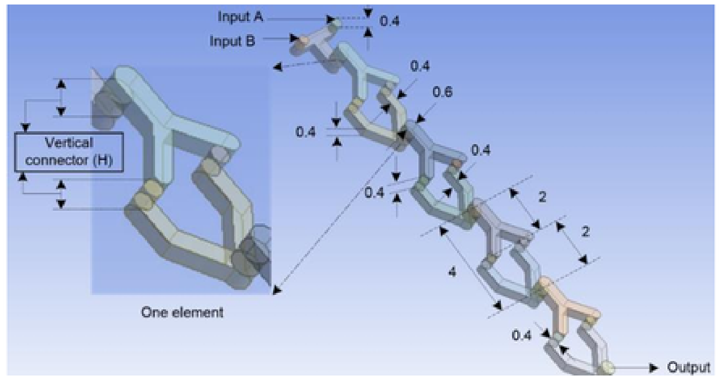
The present work numerically investigates the influence of SAR mixing unit geometry (circular vs rhombus) on mixing efficiency

Literature Review

SAR-Based Micromixers

- SAR principle enhances mixing by:
    - transverse flow, rotation
    - Interface stretching
    - Chaotic advection
- ‘Y-U’ Split and Recombine Micromixer
    - Re = 1–50
    - Mixing efficiency ≈ 90% at low Re
    - Efficiency decreases at higher Re
    - Longer flow path improves mixing
    - Increased connector height → higher pressure drop





Effect of Mixing Unit Geometry

- Shah et al. (2019)
    - Compared straight, circular, rhombus, hybrid SAR
    - Re = 0.5–100
    - Straight mixer ≈ 15% efficiency
    - Rhombus + circular (hybrid) ≈ 99% at high Re
    - Enhanced mixing due to Dean flow, secondary vortices, chaotic advection

Key Observations from Literature

- Mixing strongly depends on channel curvature, mixing unit shape, Reynolds number
- Improved mixing often comes with increased pressure drop and higher energy cost

Identified Research Gap

- Limited systematic comparison of circular vs rhombus SAR geometries under identical numerical conditions
- Need for controlled numerical evaluation across Reynolds number range

Motivation & Objectives

Motivation

- Efficient micromixing is critical for lab-on-chip systems, biomedical diagnostics, microreactors
- In laminar microflows: mixing is diffusion-dominated; efficiency decreases at higher Reynolds numbers
- SAR micromixers improve mixing, but performance strongly depends on geometry; enhanced mixing often increases pressure drop
- Optimal balance between efficiency and cost remains unclear

Objectives of the Present Work

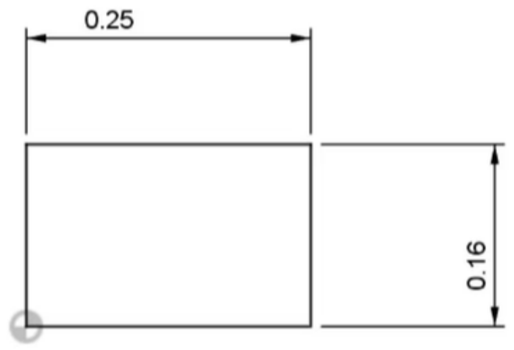
- To numerically investigate SAR micromixers with different mixing unit geometries: Circular units, Rhombus units, Baseline straight configuration
- To evaluate performance over a range of Reynolds numbers in terms of: Mixing index, Pressure drop, Secondary flow structure
- To analyze the trade-off between mixing efficiency
- To identify the most effective geometry for enhanced passive micromixing

This numerical study provides a controlled evaluation of geometry-driven mixing mechanisms and their hydraulic penalties, enabling evidence-based geometry selection for passive micromixer design.

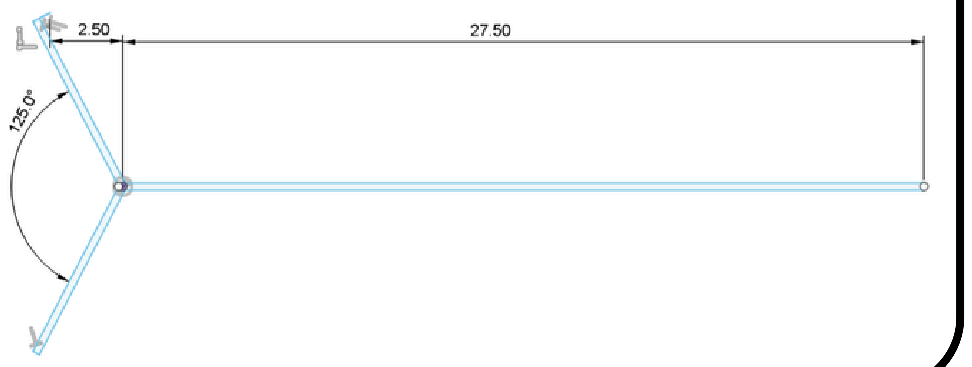
Methodology

Dimensions

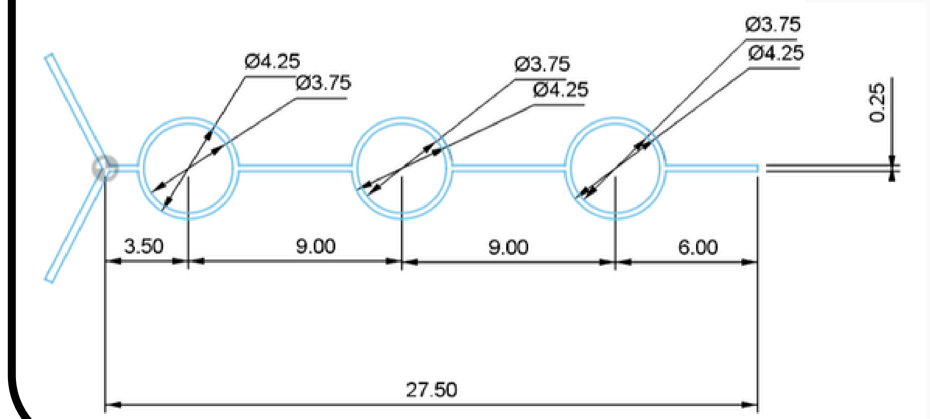
- h/w ratio = 0.65
- Width(w) = 250 µm
- Height(h) = 160 µm



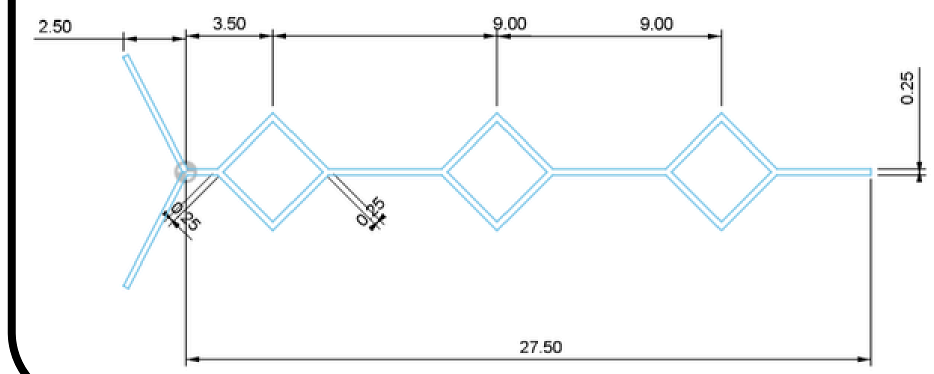
Geometry 1

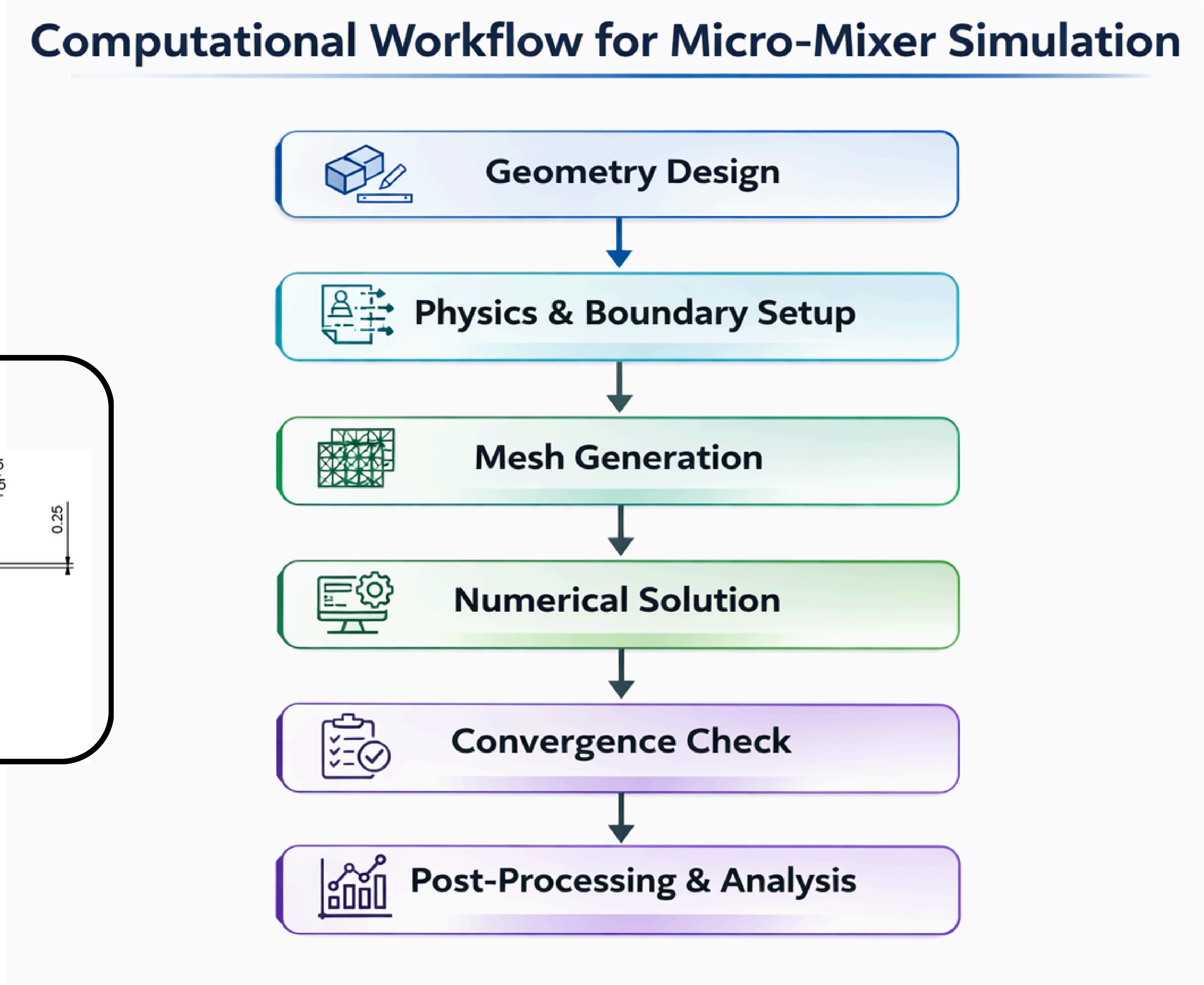


Geometry 2



Geometry 3





Methodology

Other Necessary Parameters

- Diffusion Coefficient (D) = 1×10^{-9} m²/s
- Density (ρ) = 10^3 kg/m³
- Dynamic Viscosity (μ) = 0.001 Pa·s

Equations Used

- Continuity Equation (Steady Equation)
- Navier–Stokes Equation
- Convection–Diffusion Equation
- Reynolds Number Equation

Boundary Conditions

- At Inlets:
  - Velocity boundary condition applied
  - Inlet 1 concentration = 0 mol/m³
  - Inlet 2 concentration = 1 mol/m³
- At Outlet:
  - Pressure = 0 (Gauge pressure)
- At Walls:
  - No-slip boundary condition
  - Velocity = 0 at all walls

Software Used

- ANSYS 2025R2
- Algorithm: SIMPLE
- CAD: fusion360

Some Other Formulae used

- Mixing Index $M.\,I = 1-\sqrt{\frac{\sigma^2}{\sigma^2_{max}}}$
- Standard Deviation (Variance Term) $\sigma = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(a_i-b)^2}$
- Maximum Variance Formula $\sigma_{max} = \sqrt{b(1-b)}$

σ_{max} = 0.5

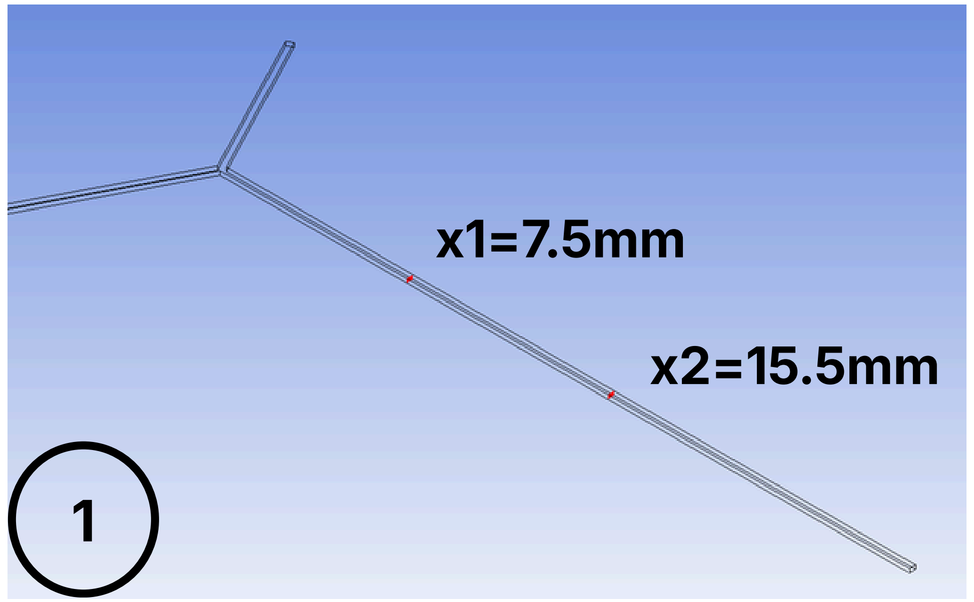
(optimal mixing value) b = 0.5

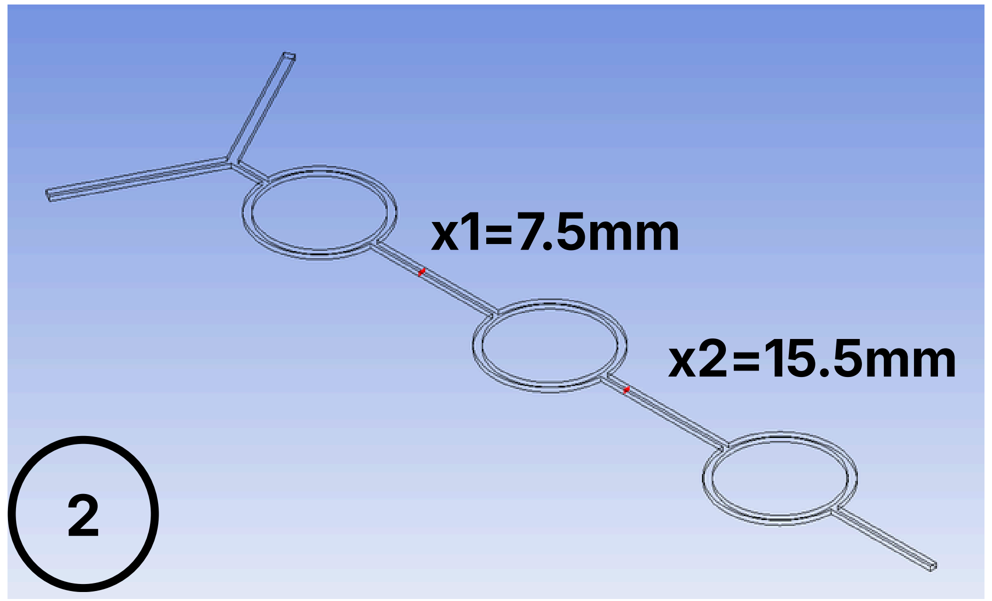
Data Collection and Calculation Method

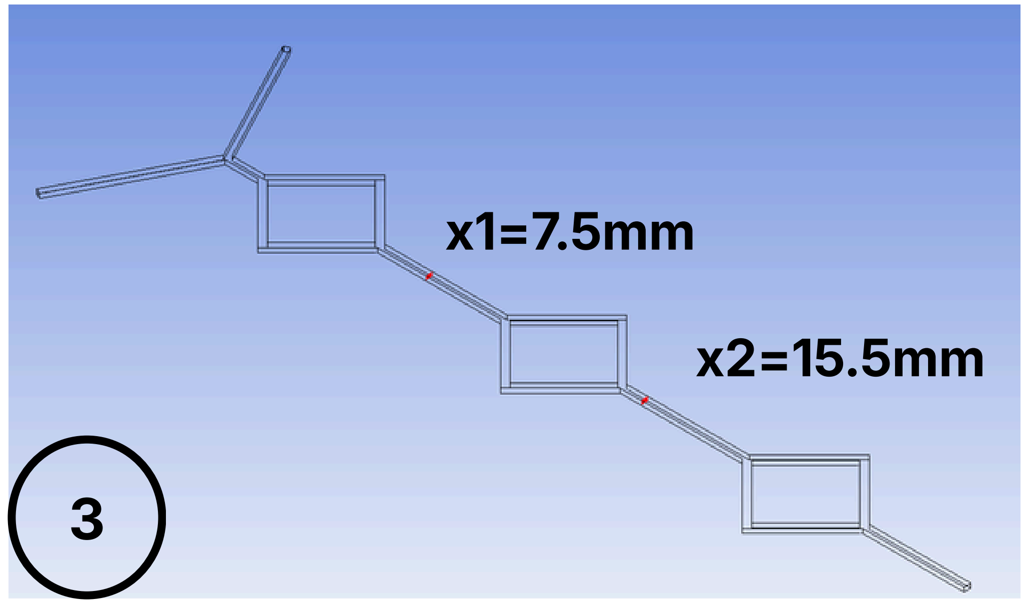
REYNOLDS NUMBER	HEIGHT	WIDTH	HYDRAULIC DIAMETER	VELOCITY [M/S]
1	160	250	195.122	0.005125
1.5	160	250	195.122	0.0076875
2	160	250	195.122	0.01025
2.5	**160**	**250**	**195.122**	**0.0128125**
3	160	250	195.122	0.015375

Accuracy is very important in evaluating mixing performance that is why 400 equally spaced sample points are taken from the cut plane at the outlet for calculating the mixing index for each of the above cases

Results and Discussions

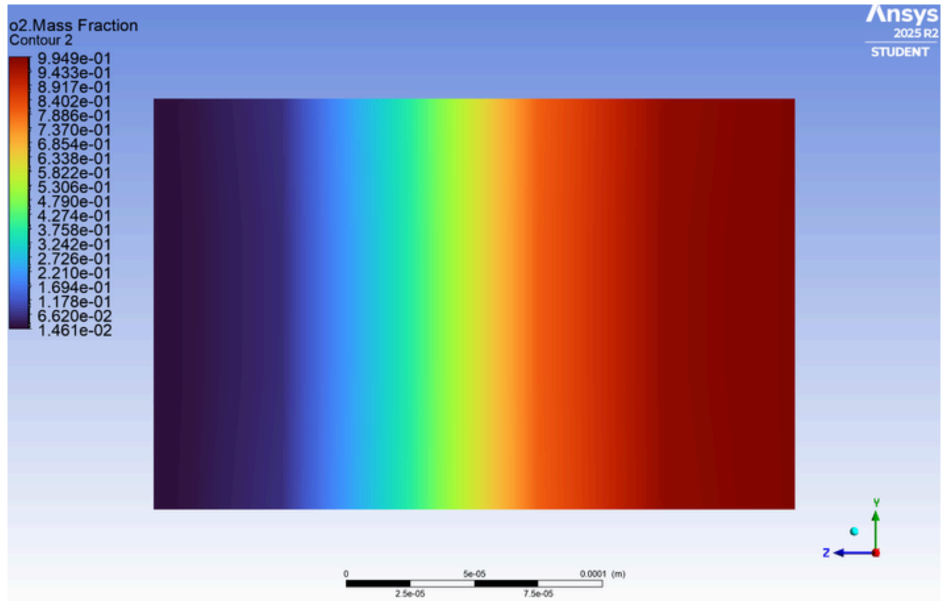






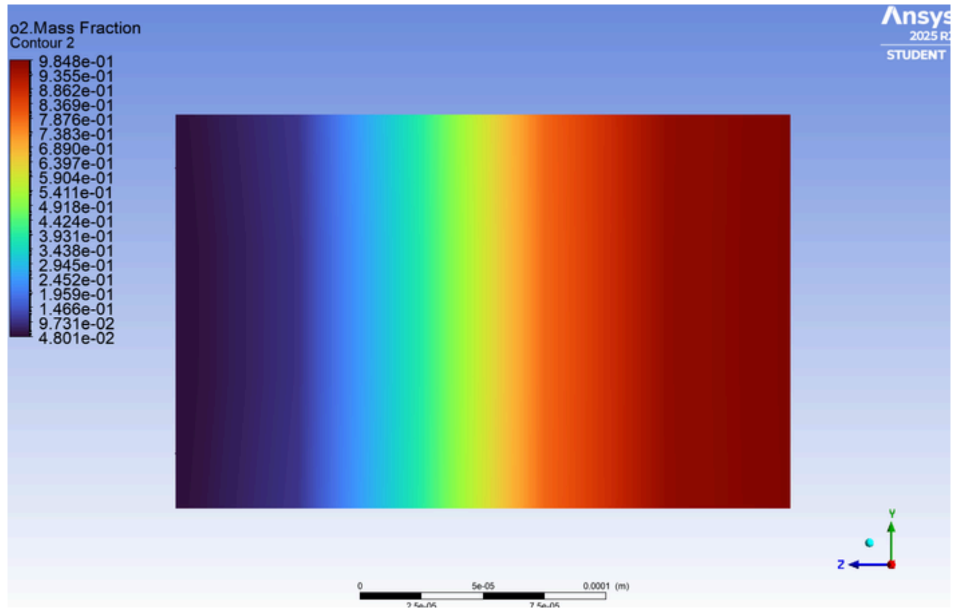
- **CONCENTRATION GRAIDENTS (Re=2.5)**

x1=7.5mm



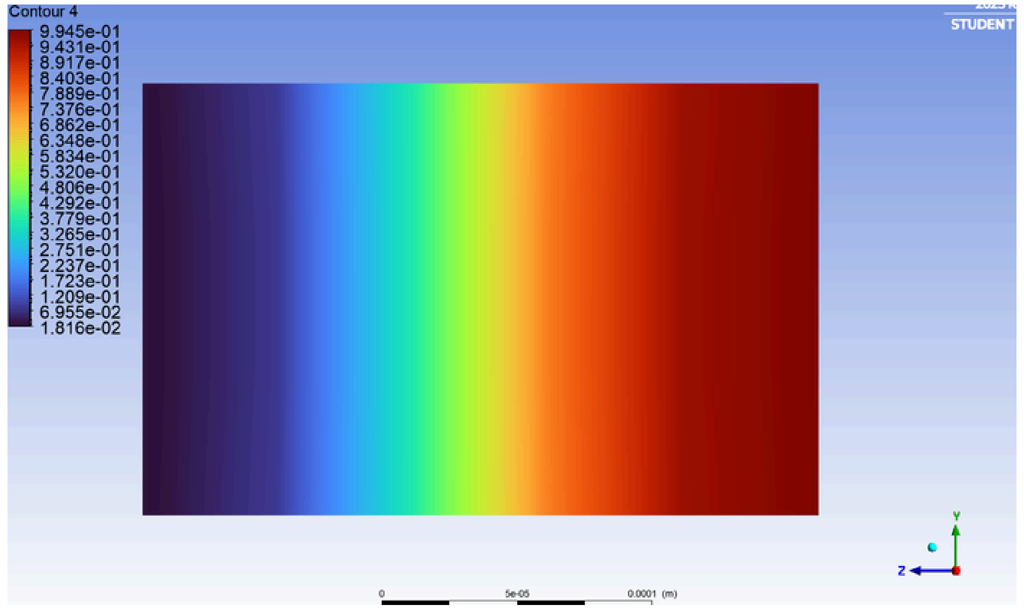
MIN. MASS FRACTION= 0.014

MAX. MASS FRACTION= 0.995



MIN. MASS FRACTION= 0.048

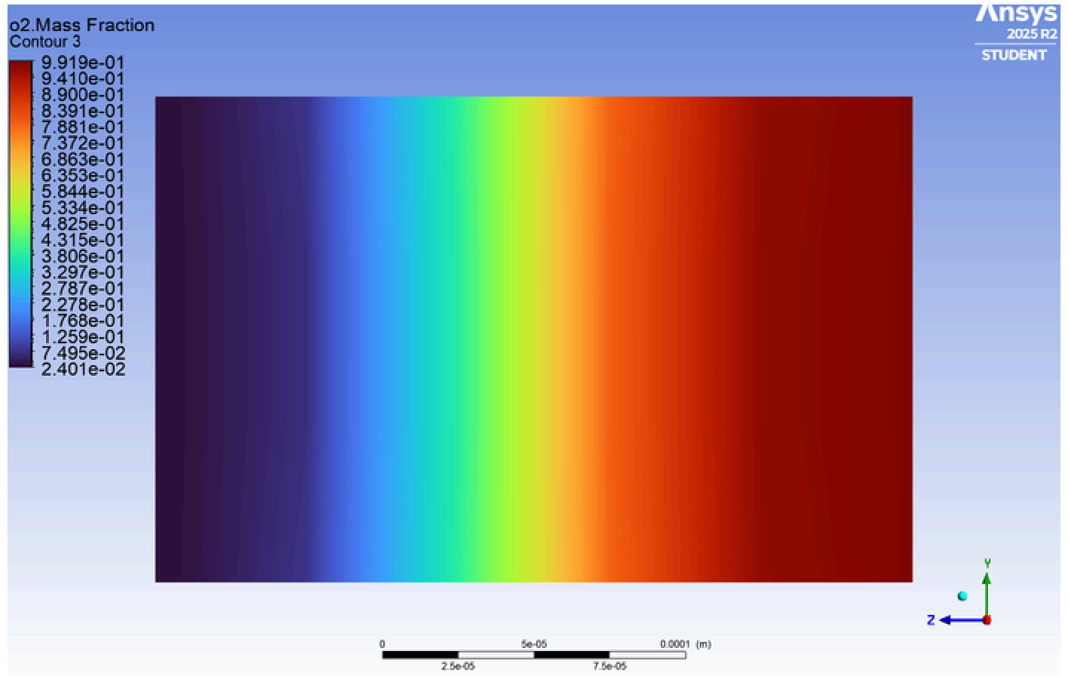
MAX. MASS FRACTION = 0.985

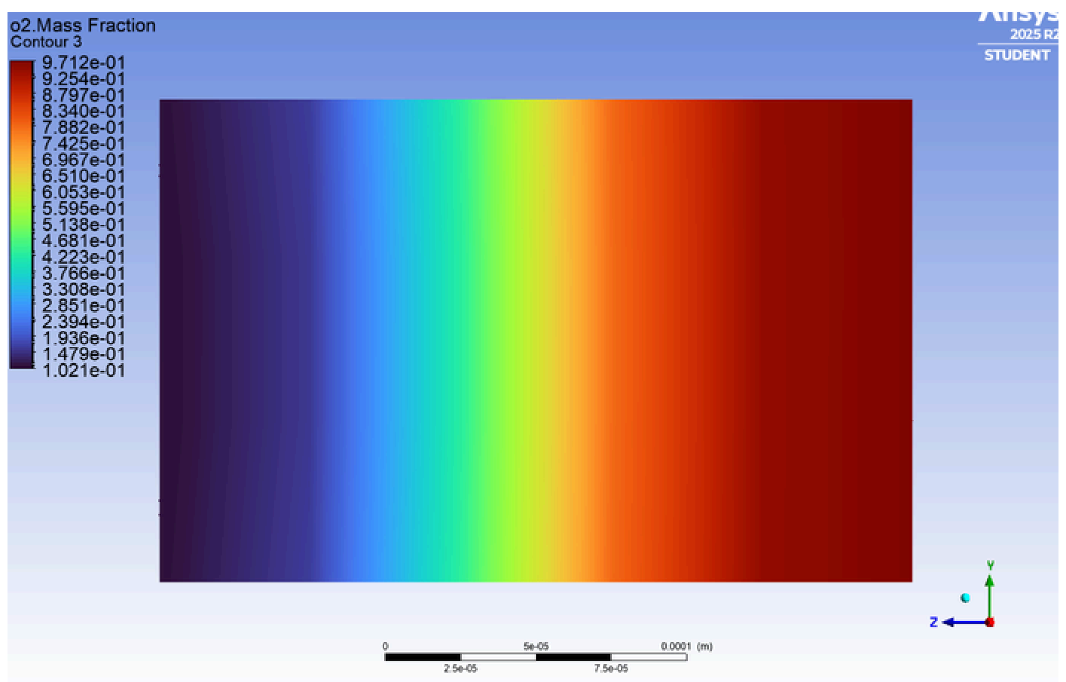


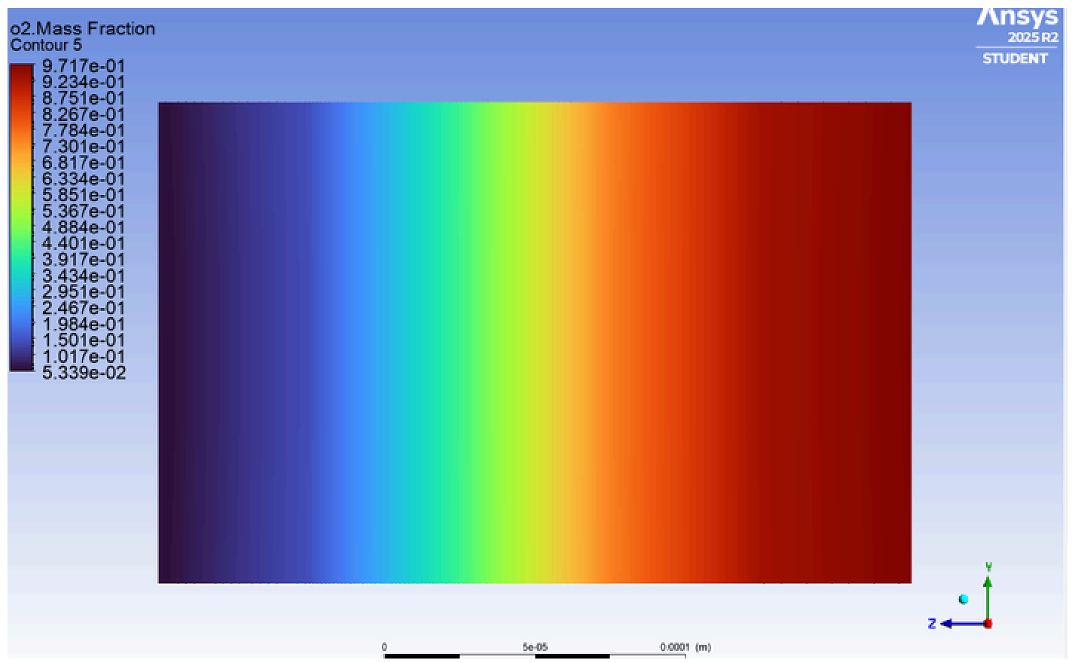
MIN. MASS FRACTION= 0.018

MAX. MASS FRACTION= 0.993

x2=15.5mm







MIN. MASS FRACTION= 0.024

MAX. MASS FRACTION = 0.991

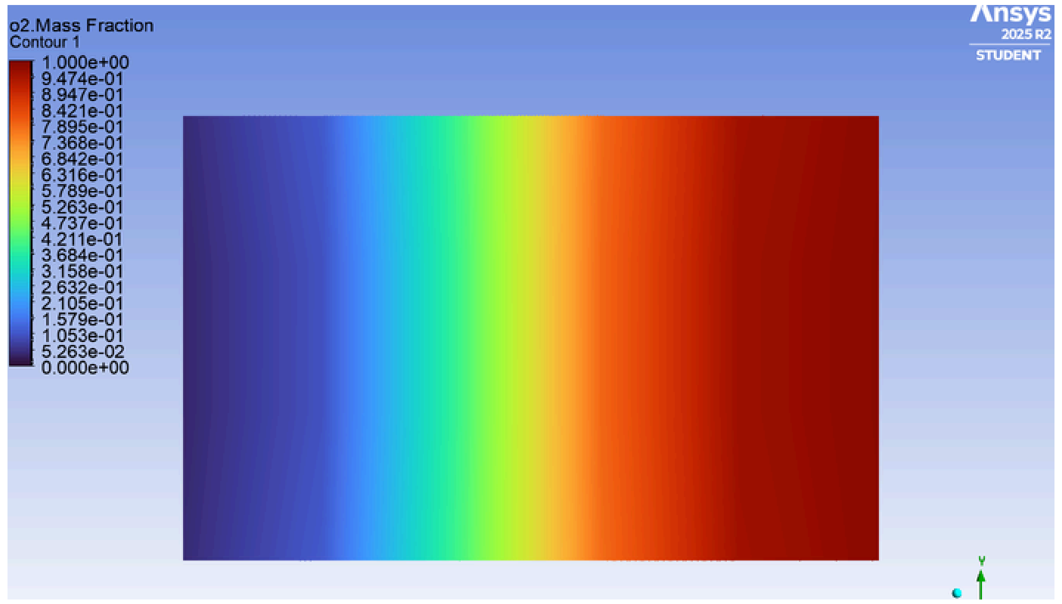
MIN. MASS FRACTION=0.103

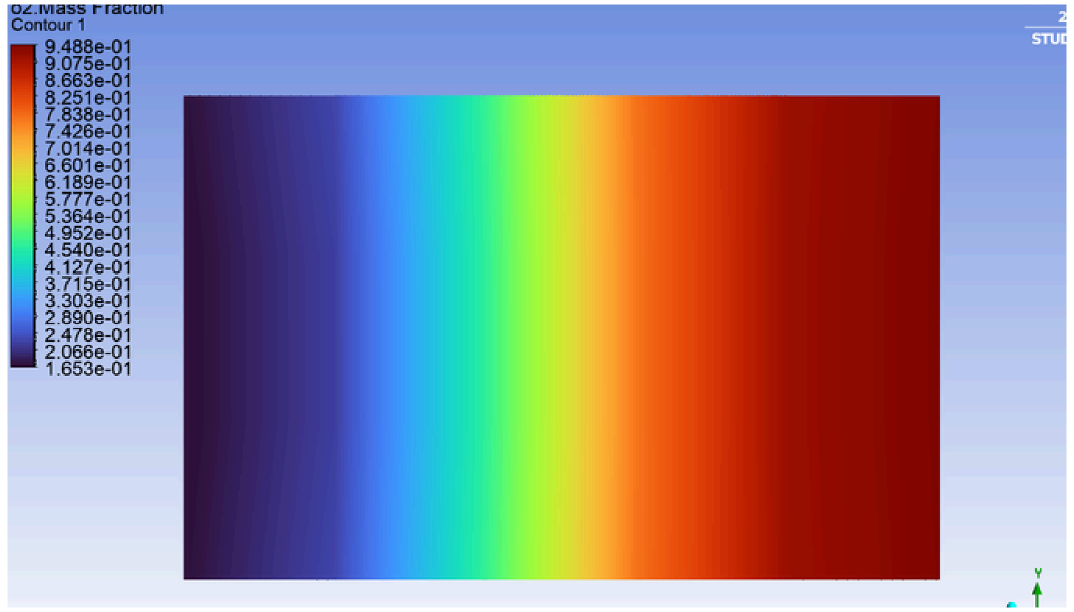
MAX. MASS FRACTION=0.971

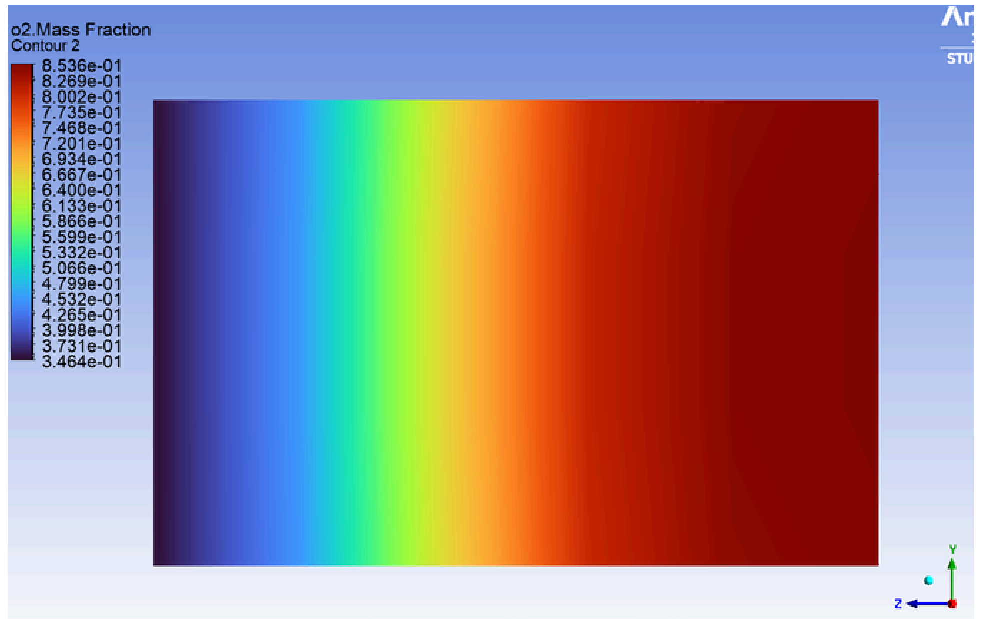
MIN. MASS FRACTION= 0.097

MAX. MASS FRACTION= 0.951

x3=outlet







MIN. MASS FRACTION= 0.037

MAX. MASS FRACTION= 0.987

MIN. MASS FRACTION= 0.165

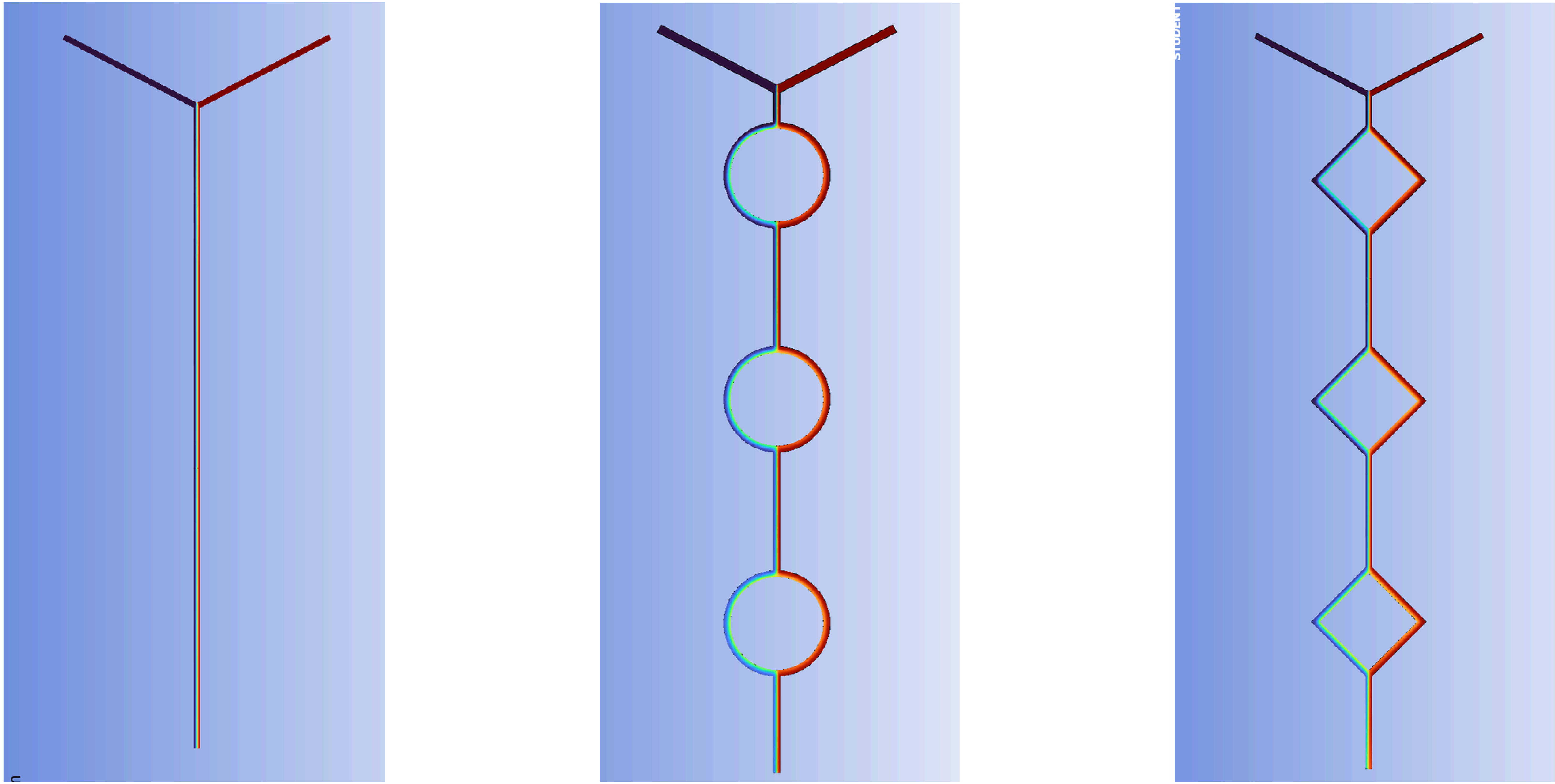
MAX. MASS FRACTION= 0.948

MIN. MASS FRACTION= 0.346

MAX. MASS FRACTION = 0.854

CONCENTRATION GRAIDENTS (Re=2.5)

Across the geometry

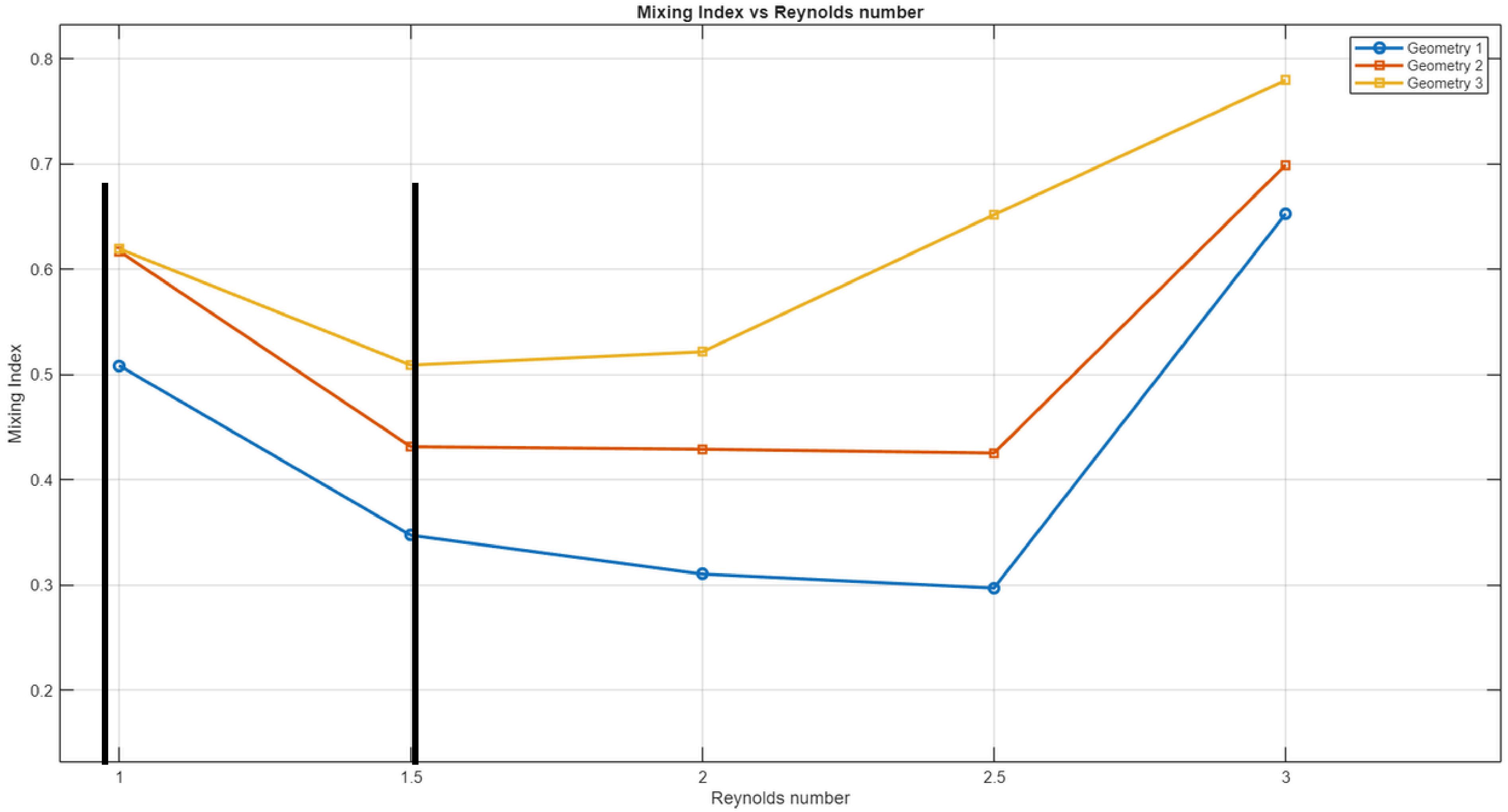


Comparative analysis

Variation of Mixing Index of each geometry with Reynolds number of the flow

At Low Reynolds Number (Re = 1 to 1.5)

- Flow is strongly laminar and diffusion-dominated.
- Fluid layers move smoothly.
- Mixing happens mainly due to molecular diffusion.
- Since velocity is low, residence time is high, so it has better mixing.
- So the mixing index starts relatively high

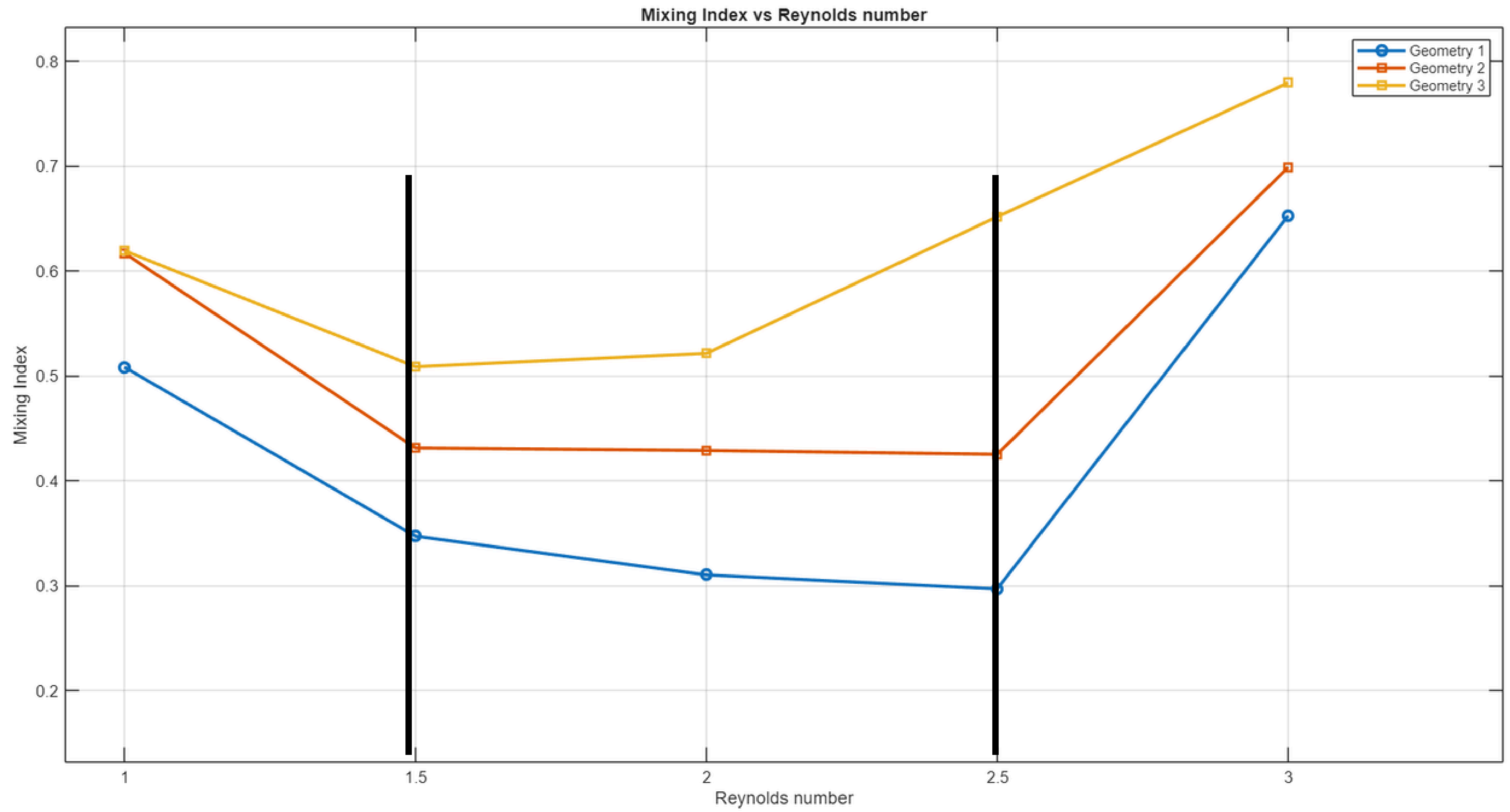


Comparative analysis

Variation of Mixing Index of each geometry with Reynolds number of the flow

<u>As Reynolds Number Increases (Re = 1.5 to 2.5)</u>

- Flow becomes more convection-dominated but still laminar.
- Velocity increases.
- Residence time decreases.
- Molecular Diffusion has less time to act.
- But the flow is still not strong enough to create vortices or chaotic advection.
- So Mixing becomes worse
- Mixing index decreases
- This region often shows minimum mixing efficiency.

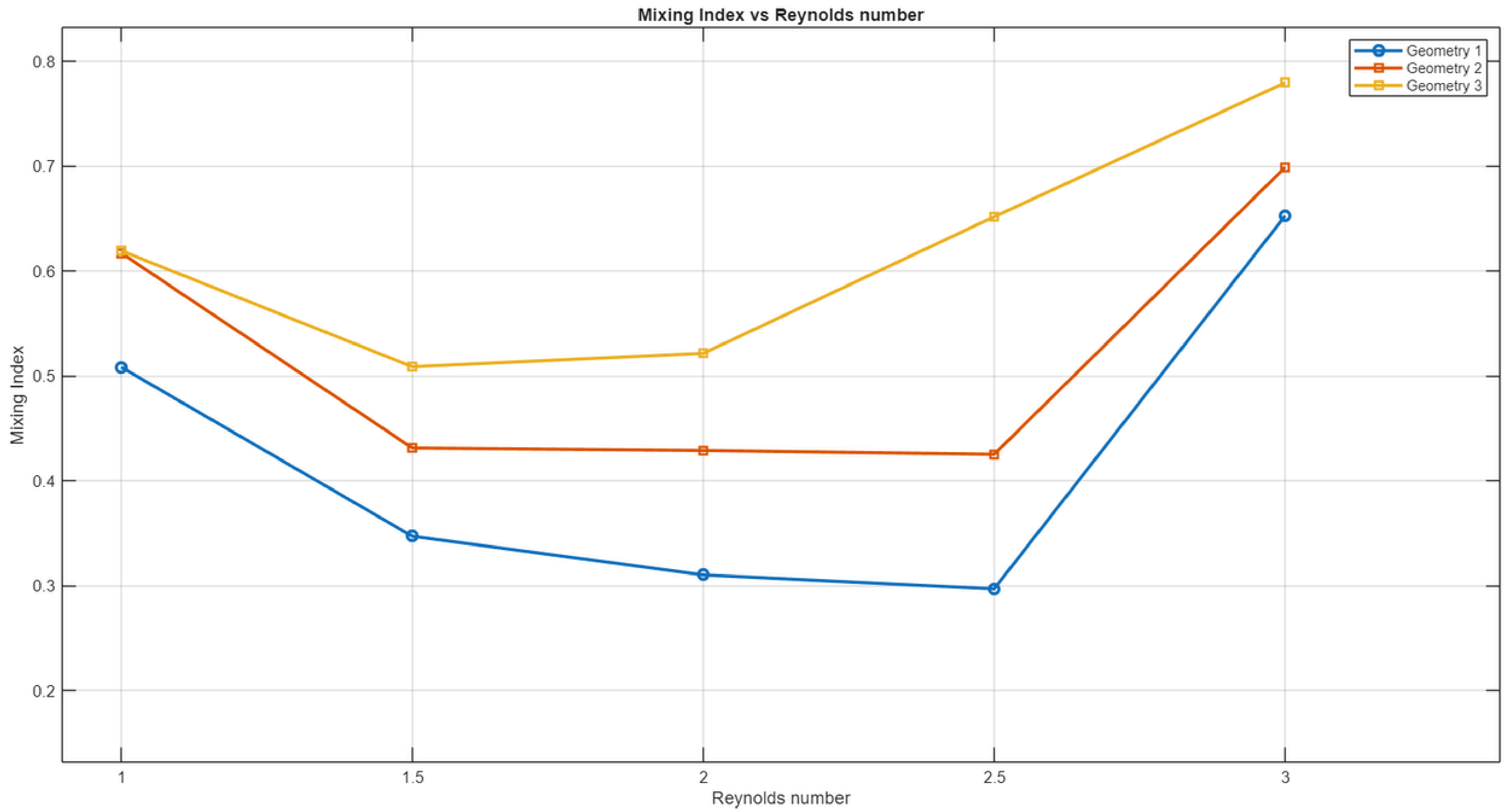


Comparative analysis

Variation of Mixing Index of each geometry with Reynolds number of the flow

<u>At Higher Reynolds Number (Re = 3)</u>

- Now inertial effects become significant and advection dominates molecular diffusion
- Even if the flow is technically laminar, you may get:
    a. Flow recirculation
    b. Chaotic advection
- These effects
    c. Enhance transverse transport
    d. Promote convective mixing
    e. So the mixing index increases sharply
    f. Transverse flow.



Conclusion

A rigorous comparative CFD study was conducted on three microchannel geometries — straight baseline, circular SAR units, and rhombus SAR units — across Re = 1–3 under steady laminar conditions. Governing physics were captured via the coupled Navier–Stokes and convection–diffusion equations.

Straight Geometry

Purely diffusion-dominated transport. Sharp concentration gradients persist at the outlet. **Lowest mixing index** of all geometries tested.

Circular SAR

Channel curvature induces secondary flow structures. Improved transverse mass transport yields **moderate mixing enhancement** over baseline.

Rhombus SAR

Strong split–recombine action generates chaotic advection. Achieves the **highest mixing index** among all geometries across the Re range studied.

Reynolds Number Effect on Mixing Performance

Mixing efficiency is non-monotonic with Re, revealing a critical transition regime where diffusion weakens before inertial effects become significant. Understanding this window is essential for optimal micromixer operation.

Key Overall Findings

Mixing performance is **strongly geometry-dependent**. Rhombus SAR units provide maximum enhancement. A **critical Re region** with minimum mixing efficiency exists..

Low Re (1–1.5)	High residence time favors diffusion. Relatively good mixing despite low inertia. Diffusion-dominated regime.
Intermediate Re (1.5–2.5)	Diffusion time reduced; inertial vortices not yet established. **Minimum mixing efficiency** — the critical trough.
Higher Re (3)	Inertial effects become significant. Recirculation zones and chaotic advection drive a **sharp recovery in mixing index**.

Future Directions

The present study establishes a solid numerical baseline. Five targeted research directions will systematically extend these findings toward a comprehensive, experimentally validated, and application-ready micromixer design framework.

Hydraulic Performance

Quantify pressure drop penalties. Define a **Performance Evaluation Criterion (PEC)** to rigorously balance mixing gain against pumping cost.

Extended Re Study

Probe $Re < 1$ (pure diffusion) and $Re > 10$ (strong inertia) to map the full operating envelope and identify the **optimal working window**.

Secondary Flow Analysis

Compute vorticity fields, evaluate the **Dean number**, and apply chaotic mixing metrics (e.g., Lyapunov exponents) for quantitative secondary flow characterization.

Parametric Optimization

Systematically vary rhombus angle, curvature radius, number of SAR units, and aspect ratio h/w to identify globally optimized micromixer geometries.

Advanced Extensions

Incorporate transient and pulsatile flow, non-Newtonian fluids, experimental validation via microfabrication, and **AI-driven geometry optimization**.

References

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